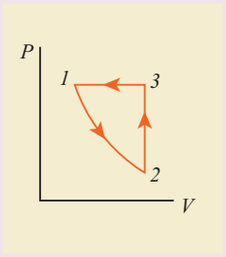
9. In a petrol engine, (internal combustion engine) air at atmospheric pressure and temperature of 20°C is adiabatically compressed in the cylinder by the piston to 1/8 of its original volume. Calculate the temperature of the compressed air.

(For air $\gamma = 1.4$)

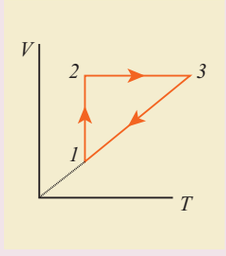
Answer: T ≅ 400°C

10. Consider the following cyclic process consist of isotherm, isochoric and isobar which is given in the figure.



Draw the same cyclic process qualitatively in the V-T diagram where T is taken along x direction and V is taken along y-direction. Analyze the nature of heat exchange in each process.

Answer:



Process 1 to 2 = increase in volume. So heat must be added.

Process 2 to 3 = Volume remains constant. Increase in temperature. The given heat is used to increase the internal energy.

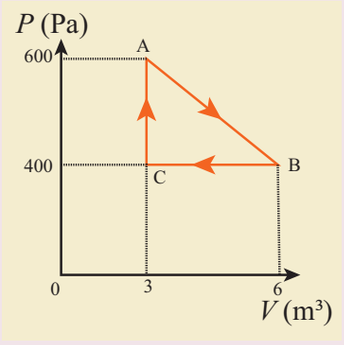
Process 3 to 1 : Pressure remains constant. Volume and Temperature are reduced. Heat flows out of the system. It is an isobaric compression where the work is done on the system.

11. An ideal gas is taken in a cyclic process as shown in the figure. Calculate

(a) work done by the gas.

(b) work done on the gas

(c) Net work done in the process



Answer: (a) $W = +1.5$kJ

(b) $W = -1.2$kJ

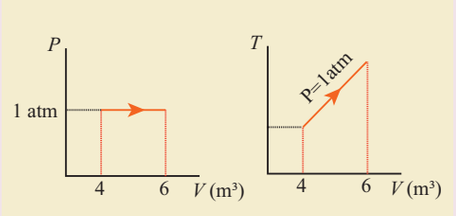
(c) $W = +300$J.

12. For a given ideal gas $6 \times 10^5 J$ heat energy is supplied and the volume of gas is increased from 4 m^3 to 6 m^3 at atmospheric pressure. Calculate (a) the work done by the gas (b) change in internal energy of the gas (c) graph this process in PV and TV diagram.

Answer: (a) $W = +202.6\ kJ$

(b) $dU = 397.4\ kJ$

(c)



13. Suppose a person wants to increase the efficiency of the reversible heat engine that is operating between 100°C and 300°C. He had two ways to increase the efficiency. (a) By decreasing the cold reservoir temperature from 100°C to 50°C and keeping the hot reservoir temperature constant (b) by increasing

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